Deep learning for multi-year ENSO forecasts

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ENSO forecast deadlock

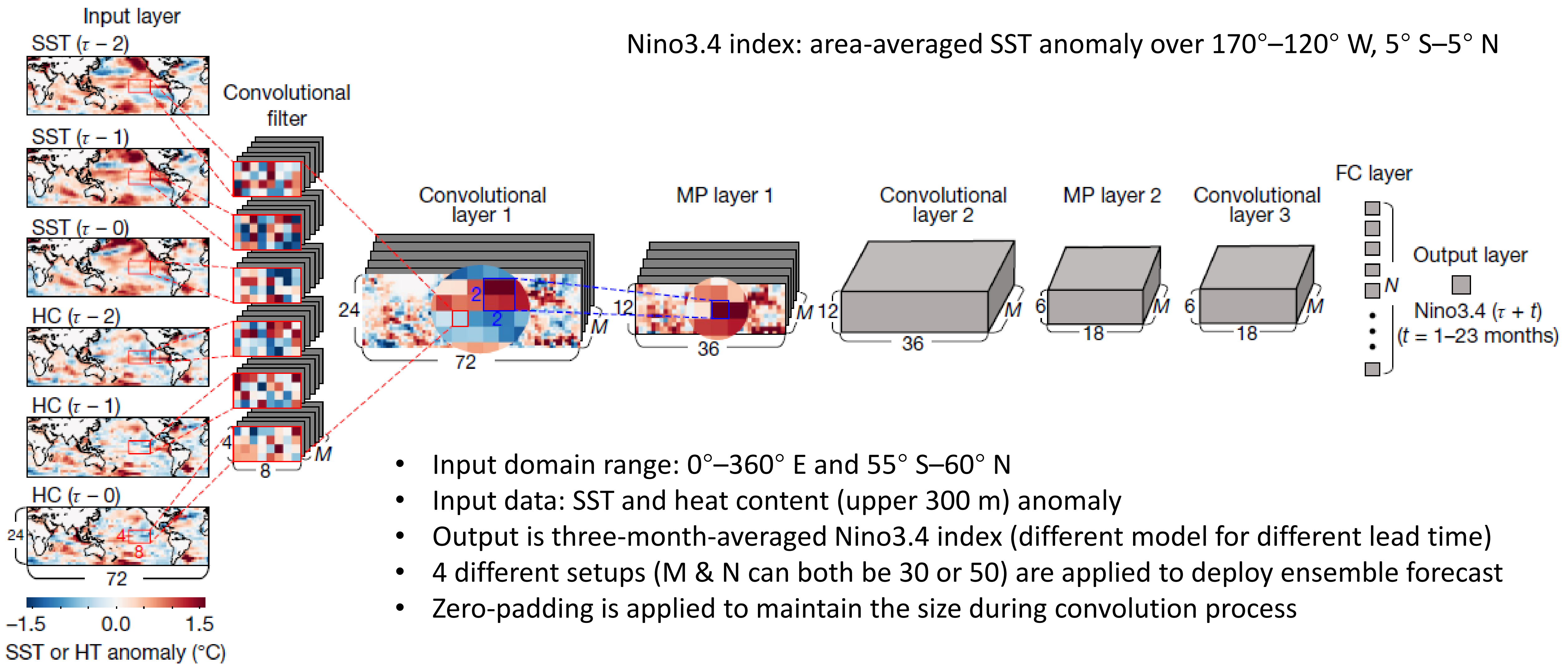
State-of-the-art dynamical forecast systems **CANNOT** make skillful prediction of ENSO for lead time **longer than a year**.

BUT:

- The presence of an oscillating element in ENSO, linked to **slowly varying oceanic variations** and their coupling to the atmosphere, suggests that multi-year forecasts are possible.
- Equatorial Pacific anomalies during several La Niña events **lingered for several years**.
- The **slowly varying component** of the equatorial winds coupled with underlying SST is predictable to some extent.
- SST anomalies outside the equatorial Pacific can lead to an ENSO event with a time-lag **longer than a year**.

It is implied that there is still room for improvement in ENSO prediction!

Deep learning setup



Nino3.4 index: area-averaged SST anomaly over 170°–120° W, 5° S–5° N

- Input domain range: 0°–360° E and 55° S–60° N
- Input data: SST and heat content (upper 300 m) anomaly
- Output is three-month-averaged Nino3.4 index (different model for different lead time)
- 4 different setups (M & N can both be 30 or 50) are applied to deploy ensemble forecast
- Zero-padding is applied to maintain the size during convolution process

Training strategy

Data shortage dilemma:

Observations of global oceanic temperature distributions are available from 1871. This means that, for each calendar month, the number of samples is **less than 150**.

Transfer learning technique:

Train the model on **similar task** with large amount of data first. Then **fine tune** the model on the target task with limited data.

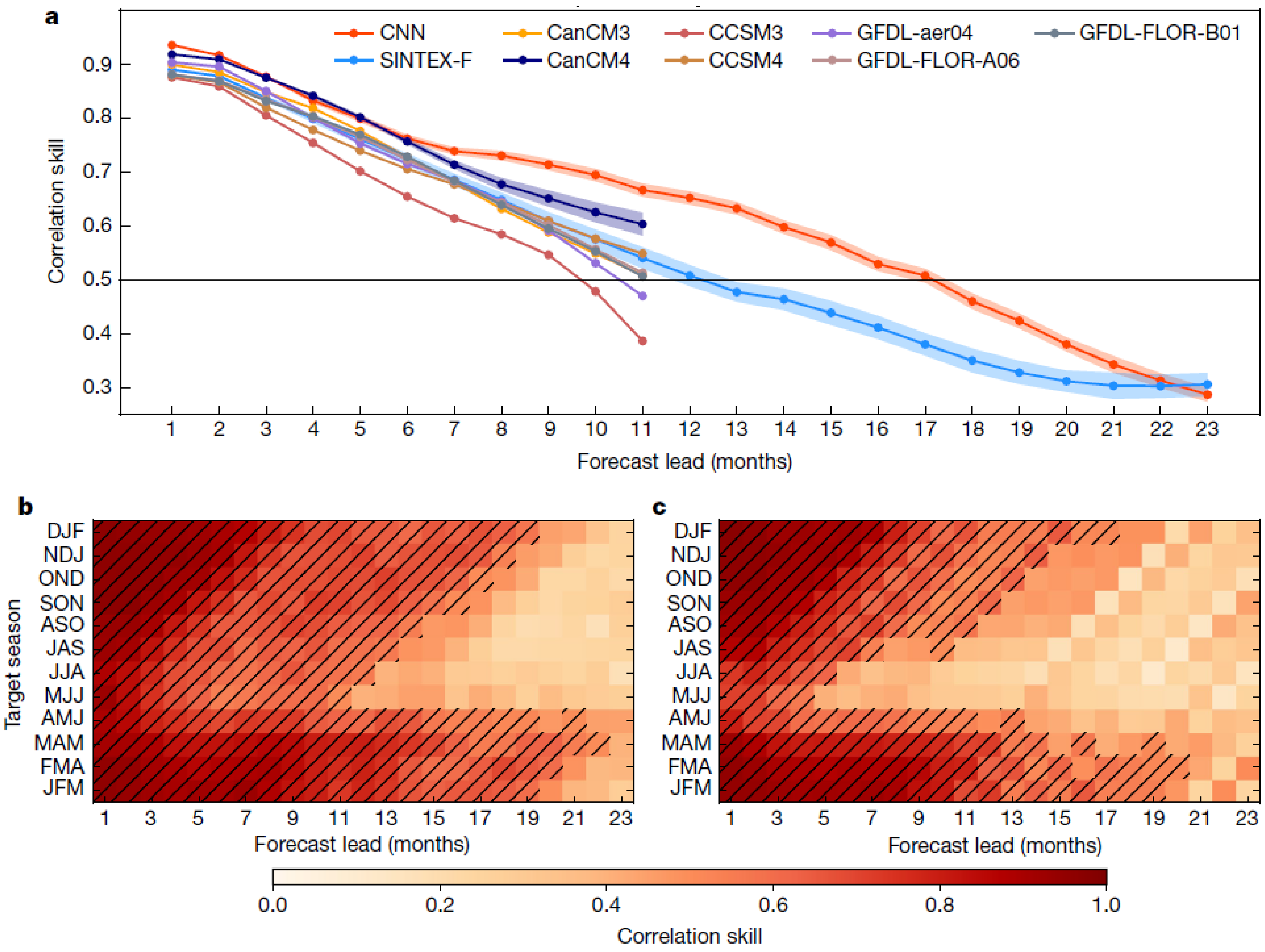
Initial model $\xrightarrow[\text{CMIP5 output}]{\text{Standard training}}$ Trained model $\xrightarrow[\text{Reanalysis data}]{\text{Fine tune}}$ Final model

Dataset details:

	Data	Period
Training dataset	CMIP5 historical run	1861-2004
	Reanalysis (SODA)	1871-1973
Validation dataset	Reanalysis (GODAS)	1984-2017

- **A ten-year gap** between the latest year in the training period and the earliest year in the validation period is leaved to remove the possible influence of **oceanic memory** in the training period on the ENSO in the validation period.
- The **systematic errors** in the CNN, reflecting those of the CMIP5 samples, are corrected after the second training period using reanalysis.
- The year in the CMIP5 models is solely dependent on the prescribed greenhouse gas forcing, so **no observational information** was added to the CMIP5 historical simulations.
- Take **one ensemble member** per model from all 21 models of CMIP5.

Forecast performance



$$C_l = \sum_{m=1}^{12} \frac{\sum_{y=s}^{e}(Y_{y,m}-\overline{Y}_m)((P_{y,m,l}-\overline{P}_{m,l}))}{\sqrt{\sum_{y=s}^{e}((Y_{y,m}-\overline{Y}_m))^2 \sum_{y=s}^{e}((P_{y,m,l}-\overline{P}_{m,l}))^2}}$$

The **all-season correlation skill** of the three-month-moving-averaged Nino3.4 index. The shading denotes 95% confidence interval.

The correlation skill of the Nino3.4 index targeted to each calendar month in the **CNN** model and the **SINTEX-F** dynamical forecast system. Hatching highlights the forecasts with correlation skill exceeding 0.5.

What we can see:

- The forecast skill of the Nino3.4 index in the CNN model is systematically **superior to all** state-of-the-art dynamical prediction systems at lead times **longer than six months**.
- The CNN model is **one of two models** with the best forecasting skills for the first six forecast lead months.
- The all-season correlation skill of the Nino3.4 index in the CNN model is above 0.5 for a lead of up to 17 months, whereas it is 0.37 at a lead of 17 months in the SINTEX-F.
- It is concluded that the CNN model provides a skillful forecast of ENSO events **up to 1.5 years** in advance: a result that is not possible using any of the state-of-the-art forecast systems.
- The CNN model also shows a higher correlation skill of the Nino3.4 index **for almost all targeted seasons**, compared to SINTEX-F
- The correlation skill improvement is especially robust for predictions targeting the seasons **between the late boreal spring and autumn**.
- The CNN model is **less affected** by spring predictability barriers.

Deep learning wins!

Further analysis

The skill of the FC-NN model with a **different number of predictors**

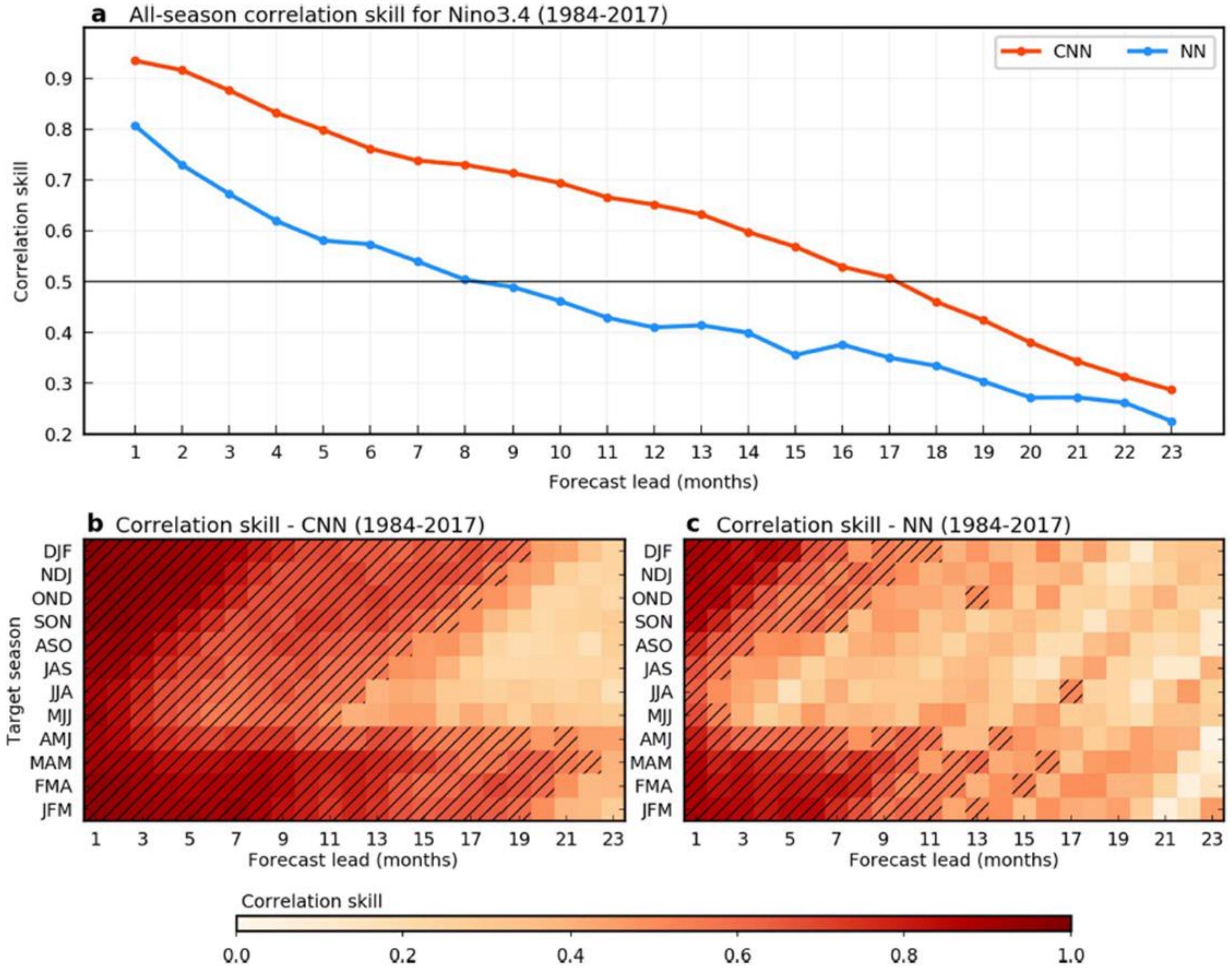
		Atlantic, North Pacific				
	Number of EOF PCs	5	6	7	8	9
Indian-Pacific	5	0.43	0.44	0.50	0.41	0.41
	6	0.42	0.43	0.50	0.41	0.40
	7	0.43	0.41	0.49	0.40	0.38
	8	0.42	0.41	0.48	0.39	0.38
	9	0.46	0.47	$\boxed{0.52}$	0.46	0.45
	10	0.21	0.30	0.38	0.23	0.39

FC-NN model setup

Input: EOF-PC of SST and heat contents in Indo-Pacific, Atlantic and North Pacific regions
Output: Nino3.4 index
Training data: reanalysis data from 1871 to 1973 & CMIP5 model output
Model structure: 2 hidden layers with 20 nodes
Activation function: hyperbolic tangent function

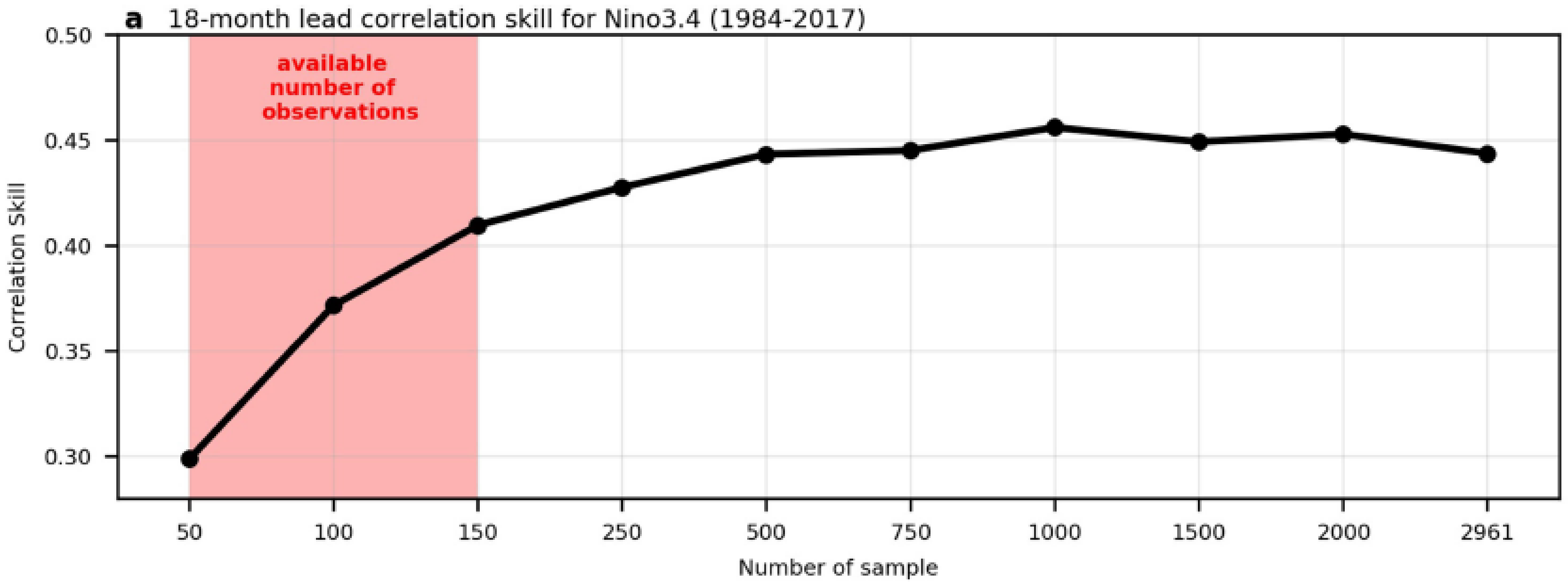
Best performance: 9 PCs in Indian-Pacific region and 7 PCs in Atlantic, North Pacific region

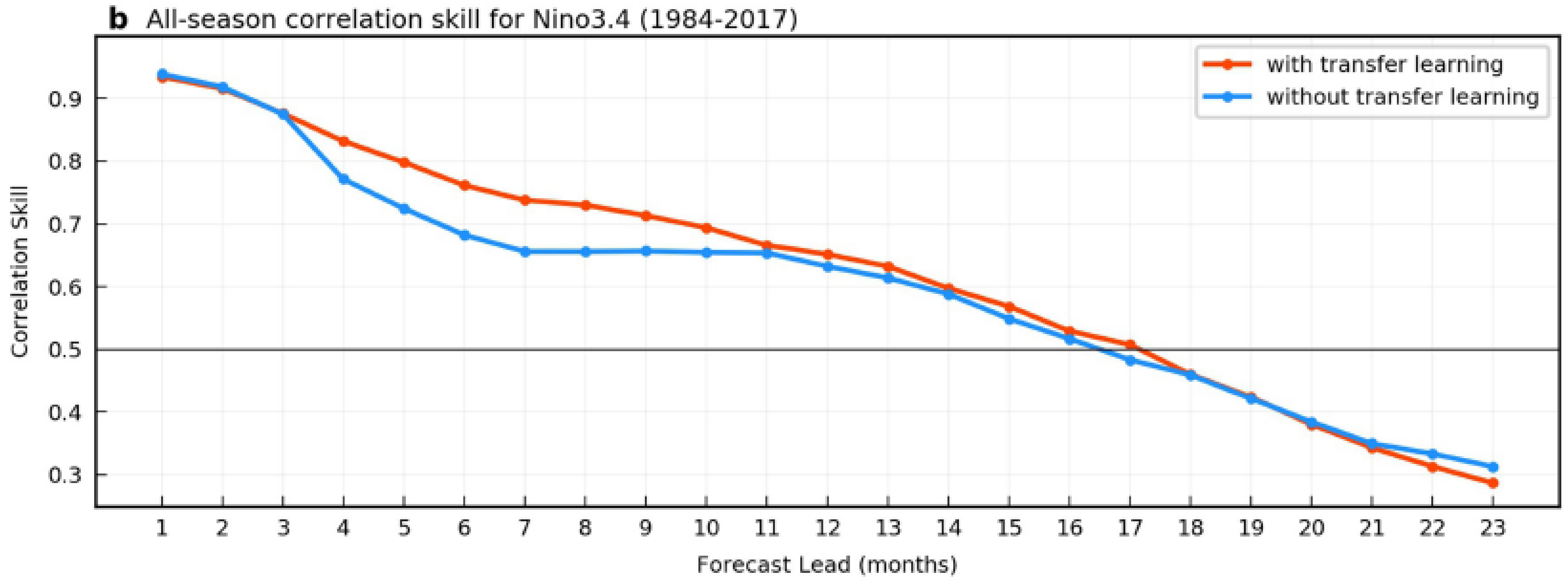
Best FC-NN model vs CNN model



The ability to predict an ENSO in the CNN is **systematically superior** to that in the neural network model.

Transfer learning effect

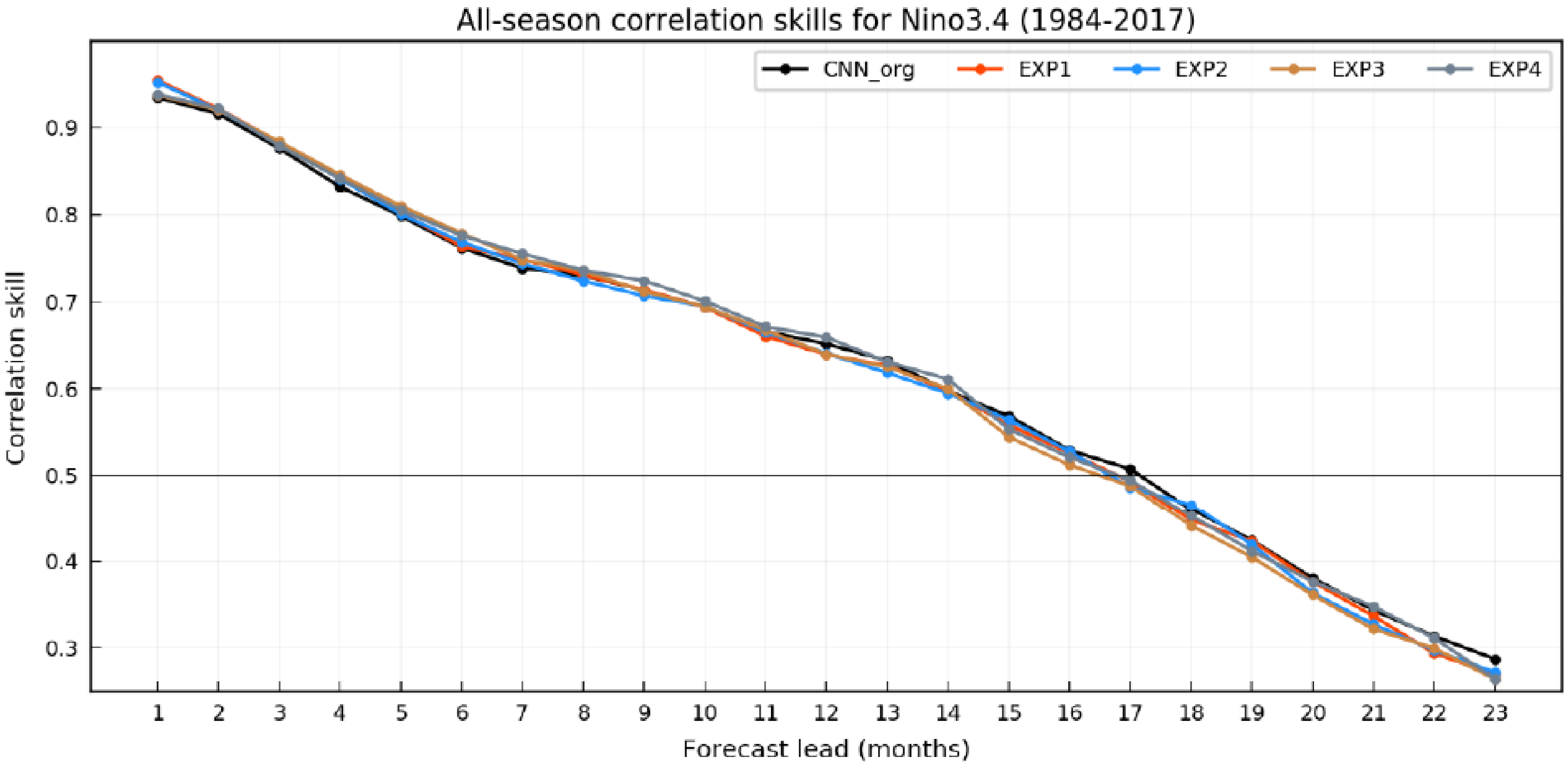




No transfer learning experiment replaces reanalysis data with corresponding CMIP5 output (the same period) to maintain the same training data size.

Transfer learning technique **contributes a lot** to the final performance.

Training dataset sensitivity



No significant reliance on training dataset arrangement

CMIP5 model data difference:

CNN_org: the first ensemble member of all CMIP5 models

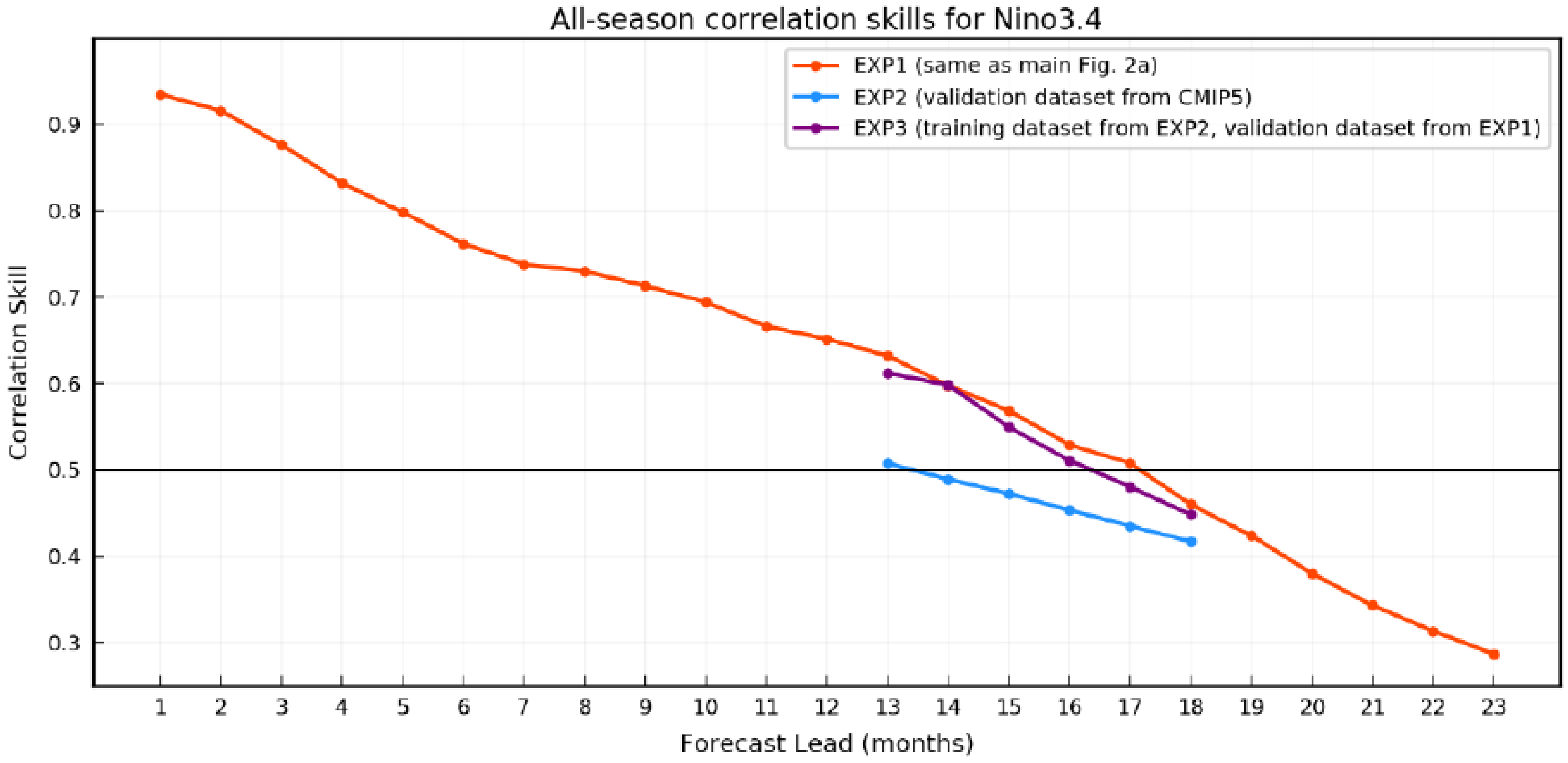
EXP1: the second ensemble member of all CMIP5 models

EXP2: randomly select a member if model provides 2 or more members

EXP3: randomly select 21 members out of all 42 members (total number same as former exps)

EXP4: the training samples are randomly selected for each year among 42 realizations

Validation dataset sensitivity

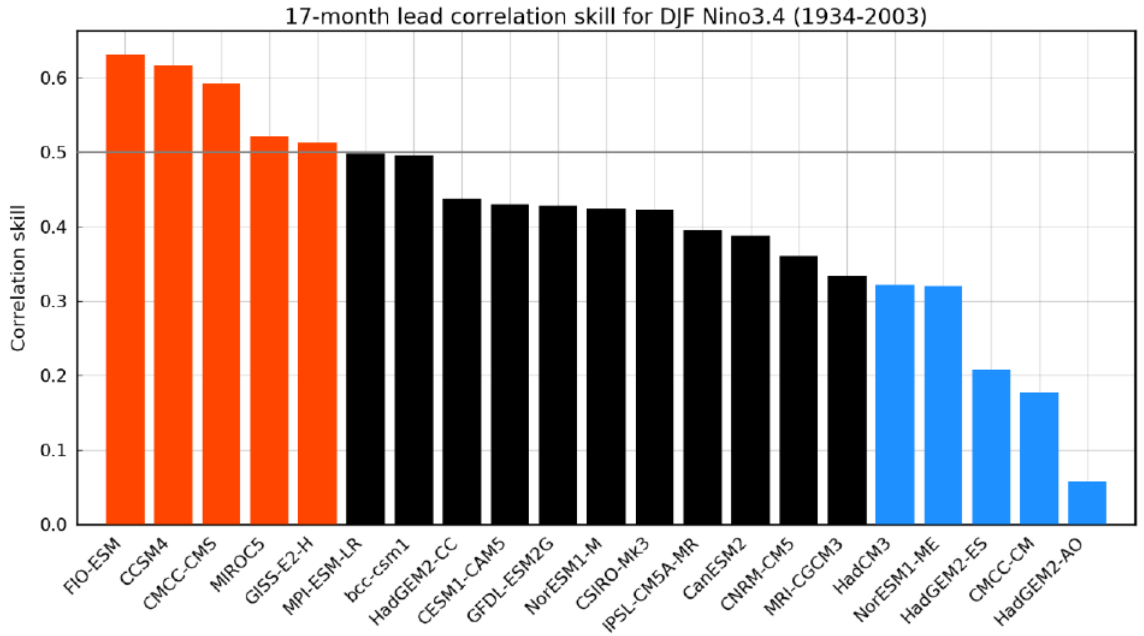


The similar correlation skill between EXP1 and EXP3 indicates that the forecast skill is **not much changed** due to the different sampling for the CNN model training.

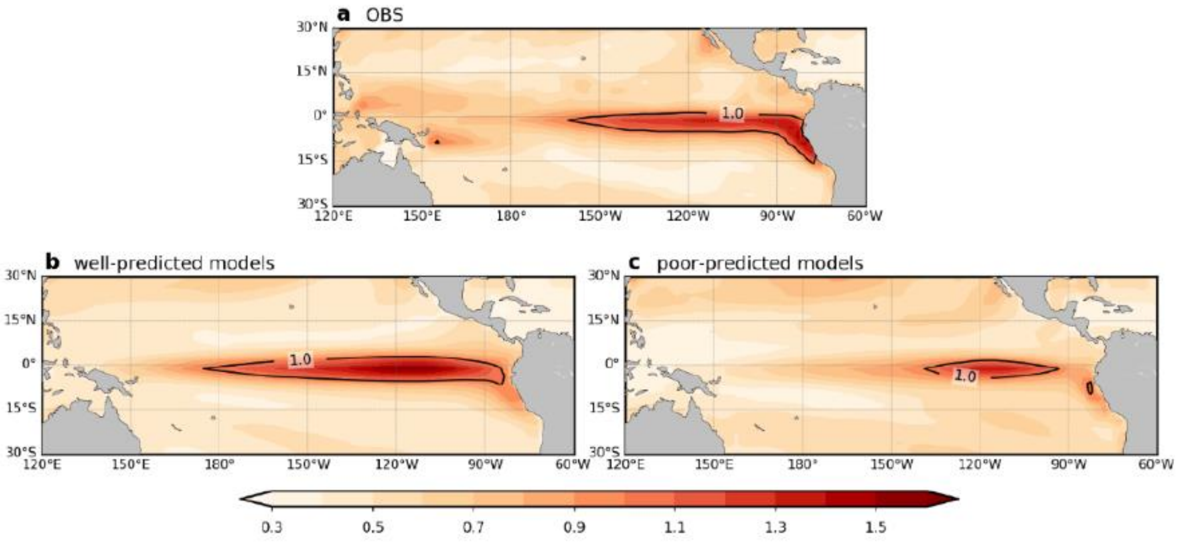
The low correlation skill in EXP2 indicates that the CNN model **has difficulty when predicting the modeled ENSO index**.

EXP1 train: All CMIP5 output + reanalysis(1871-1973), vali: reanalysis(1984-2017)
EXP2 train: First half of all CMIP5 output, vali: last half of CMIP5 output
EXP3 train: Same as EXP2, vali: same as EXP1

Prediction on model ENSO index



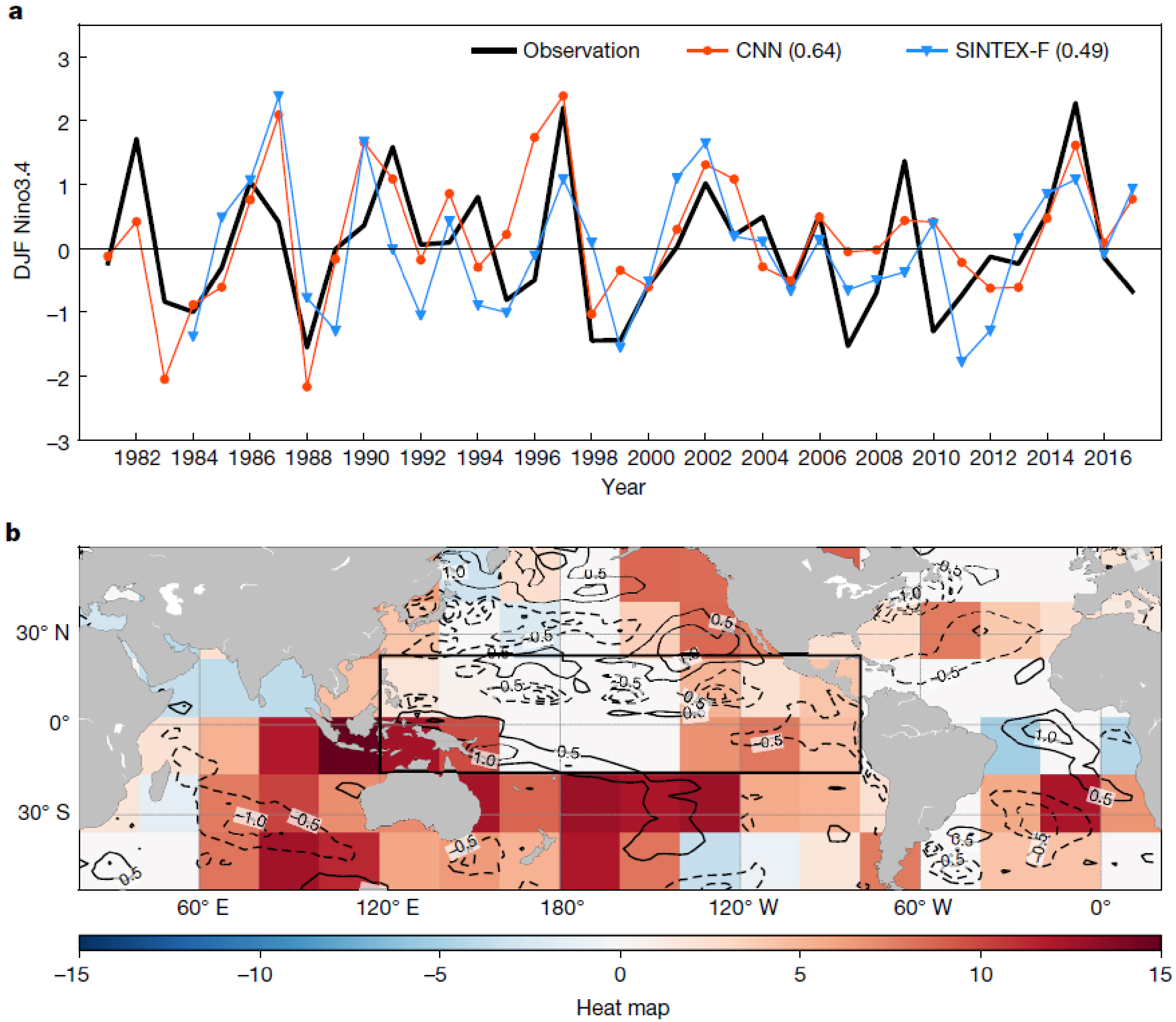
The training samples are the **first half** of the historical simulations in all CMIP5 models. While the validation set is the **last half** of different models.



The **standard deviation** of the SST anomalies during DJF season in reanalysis, well-predicted models and poor-predicted models during 1979 to 2001.

This shows that the correlation skill of the CNN model is **quite stable and robust** even predicting the modeled ENSO as far as the ENSO dynamics is simulated realistically. The validation dataset in unrealistic models is responsible for the low correlation skill in EXP2 to a large extent.

Heat map analysis



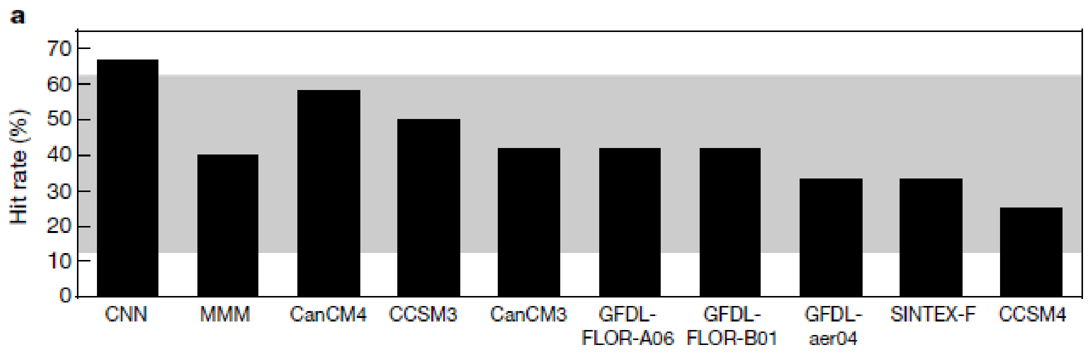
The Nino3.4 index for the DJF season for the 18-month-lead forecast demonstrates that the CNN model **correctly predicts the ENSO amplitude**. But why?

$$h^{x,y} = \sum_{n=1}^{N} \left\{ \tanh\left[\sum_{m=1}^{M_L} \left(W_{F,m,n}^{x,y} v_{L,m}^{x,y} \right) + \frac{b_{F,n}}{X_L Y_L} \right] W_{O,n} \right\} + \frac{b_O}{X_L Y_L}$$

The **heat map of the predictors** for the DJF season in 1997/98. Positive (negative) values in the heat maps denote predictors over certain regions contributing to the prediction of a positive (negative) Nino3.4. The anomalies over the **tropical western Pacific, Indian Ocean** and **subtropical Atlantic** are the main contributors to the successful prediction of the 1997/98 El Nino.

El Nino flavor forecast

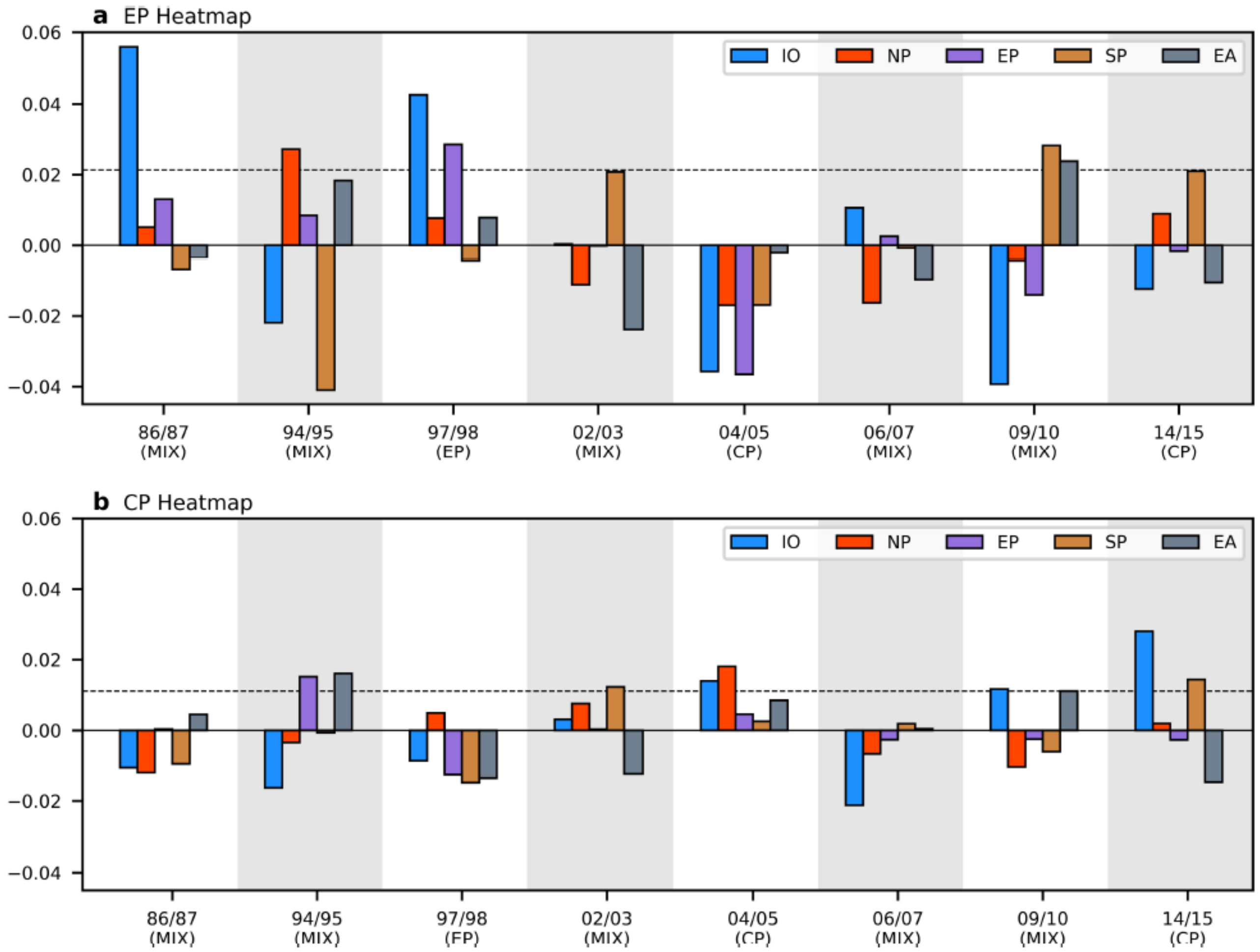
Year	OBS	CNN	SINTEX-F	CanCM3	CanCM4	CCSM3	CCSM4	GFDL-aer04	GFDL-FLOR-A06	GFDL-FLOR-B01
1976	EP	EP	-	-	-	-	-	-	-	-
1977	CP	CP	-	-	-	-	-	-	-	-
1979	MIX	MIX	-	-	-	-	-	-	-	-
1982	EP	EP	-	-	-	-	-	-	-	-
1986	MIX	MIX	MIX	MIX	MIX	MIX	MIX	EP	CP	CP
1987	MIX	CP	MIX	EP	EP	EP	CP	EP	EP	EP
1990	CP	MIX	MIX	MIX	MIX	CP	MIX	MIX	MIX	MIX
1991	MIX	CP	CP	MIX	MIX	MIX	EP	EP	EP	EP
1994	MIX	MIX	EP	EP	EP	EP	EP	EP	EP	EP
1997	EP	EP	MIX	MIX	MIX	MIX	MIX	MIX	MIX	MIX
2002	MIX	MIX	MIX	EP	MIX	MIX	EP	MIX	MIX	MIX
2004	CP	CP	MIX	EP	CP	EP	EP	EP	CP	CP
2006	MIX	MIX	MIX	MIX	MIX	MIX	MIX	MIX	MIX	MIX
2009	MIX	MIX	EP	MIX	MIX	MIX	MIX	MIX	MIX	MIX
2014	CP	CP	MIX	MIX	MIX	MIX	MIX	CP	MIX	MIX
2015	MIX	CP	CP	MIX	MIX	EP	EP	CP	MIX	MIX
Hit rate (%)	-	66.67 (75.00)	33.33	41.67	58.33	50	25	33.33	41.67	41.67

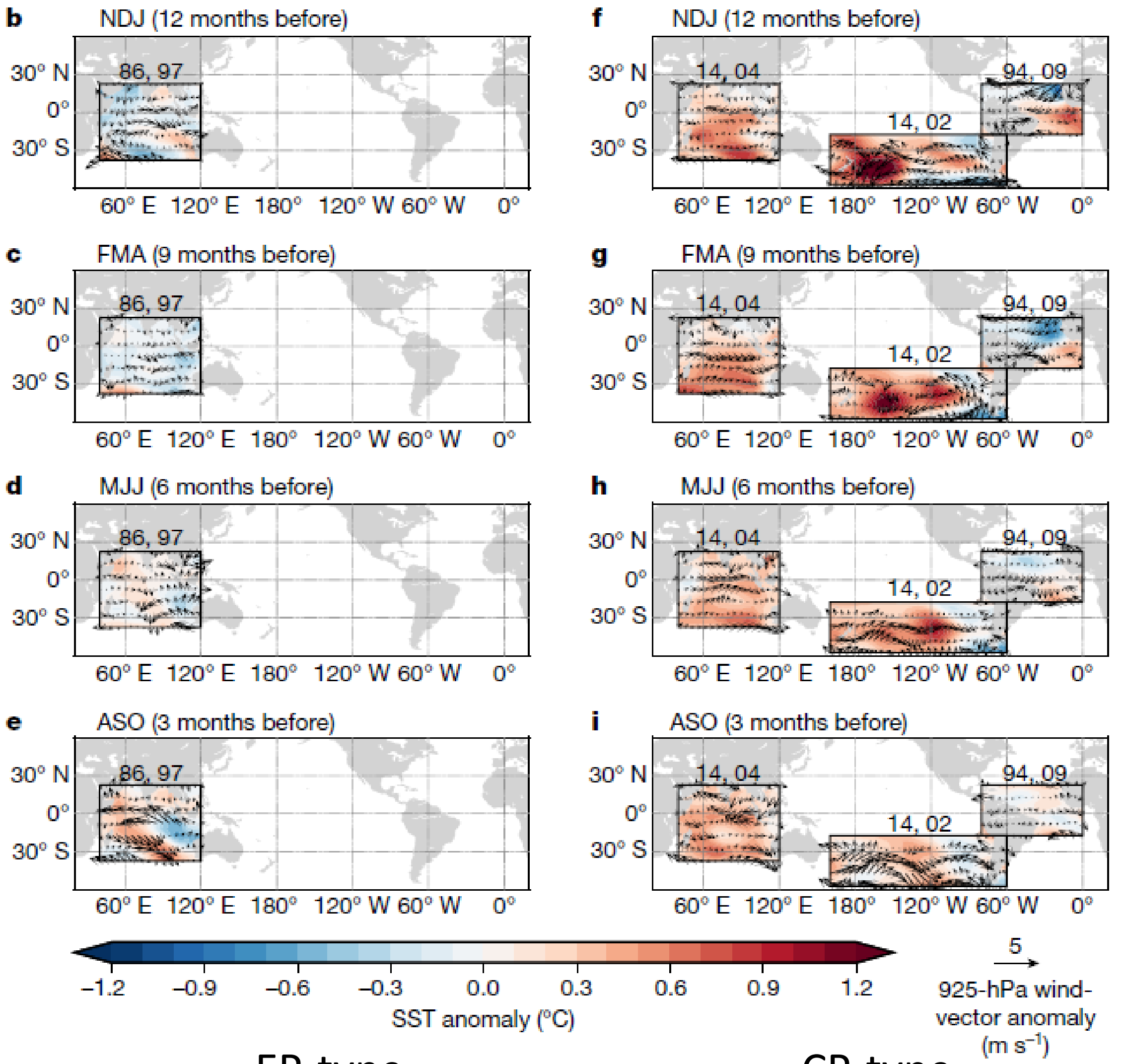


Gray zone indicates 95% confidence interval. The CNN model overcomes a long-standing weakness of the state-of-the-art forecast models

Another CNN model to predict El Nino flavor. The CNN output corresponds to the **percentage occurrence** of three El Nino categories: CP-type, EP-type and a mixture of the two. **Only the CMIP5 model outputs** is used as training data.

Heat map analysis





Area-averaged heat map value of EP-type and CP-type over south Pacific, equatorial Pacific, north Pacific, Indian Ocean, and equatorial Atlantic. First two cases having the biggest heat map value for each ocean basin are selected as the most favorable pattern for EP or CP development.

The CNN can be a powerful tool to reveal complex ENSO mechanisms!

Conclusions

- The CNN model provides a skillful forecast of ENSO events up to 1.5 years in advance: a result that is not possible using any of the state-of-the-art forecast systems.
- The skill uncertainty produced by changing the training and validation dataset is small, indicating that the CNN can provide skillful real-time forecasts.
- The CNN model is even successful in predicting the modelled ENSO index in some of the CMIP5 models that capture realistic ENSO dynamics.
- The CNN model can predict the spatial complexity of the El Nino events with great precision by skillfully predict ENSO flavor.
- The CNN model can be a powerful tool to reveal complex ENSO mechanisms.